

# Artificial Intelligence

## Course Information

- **Course Code:** CSC462 Course
- **Title:** Artificial Intelligence
- **Credit Hours:** 4 (3,1)

## Text and Reference Books

- **Textbook:** Artificial Intelligence: A Modern Approach, Russell, S., and Norvig, P., Pearson, 2020.
- **Reference Book:** Artificial Intelligence Basics: A Non-Technical Introduction, Taulli, T., Apress, 2019.



# Machine Learning Concepts: Training, Testing, Overfitting, Underfitting, and Regularization

Understanding Core Principles

# Introduction to Key Machine Learning Concepts

- Understanding training
- Understanding Testing
- Overfitting
- Underfitting
- Regularization.

# Training Data

## **Definition:**

Training is the process where a machine learning model learns from a labeled dataset by identifying patterns.

## **Key Points:**

- Training involves feeding the model with labeled data.
- The goal is for the model to generalize well to unseen data.

# Training Data

- **Purpose:** Helps the model learn patterns and make predictions.
- **Process:**
  - 1. Collect data.
  - 2. Preprocess data (cleaning, normalization).
  - 3. Train the model using this data.
- Example: Image recognition - training on labeled images.

# Testing Data

## Definition:

Testing is the process of evaluating a trained model's performance on unseen data. The purpose of testing is to determine if the model has generalized well.

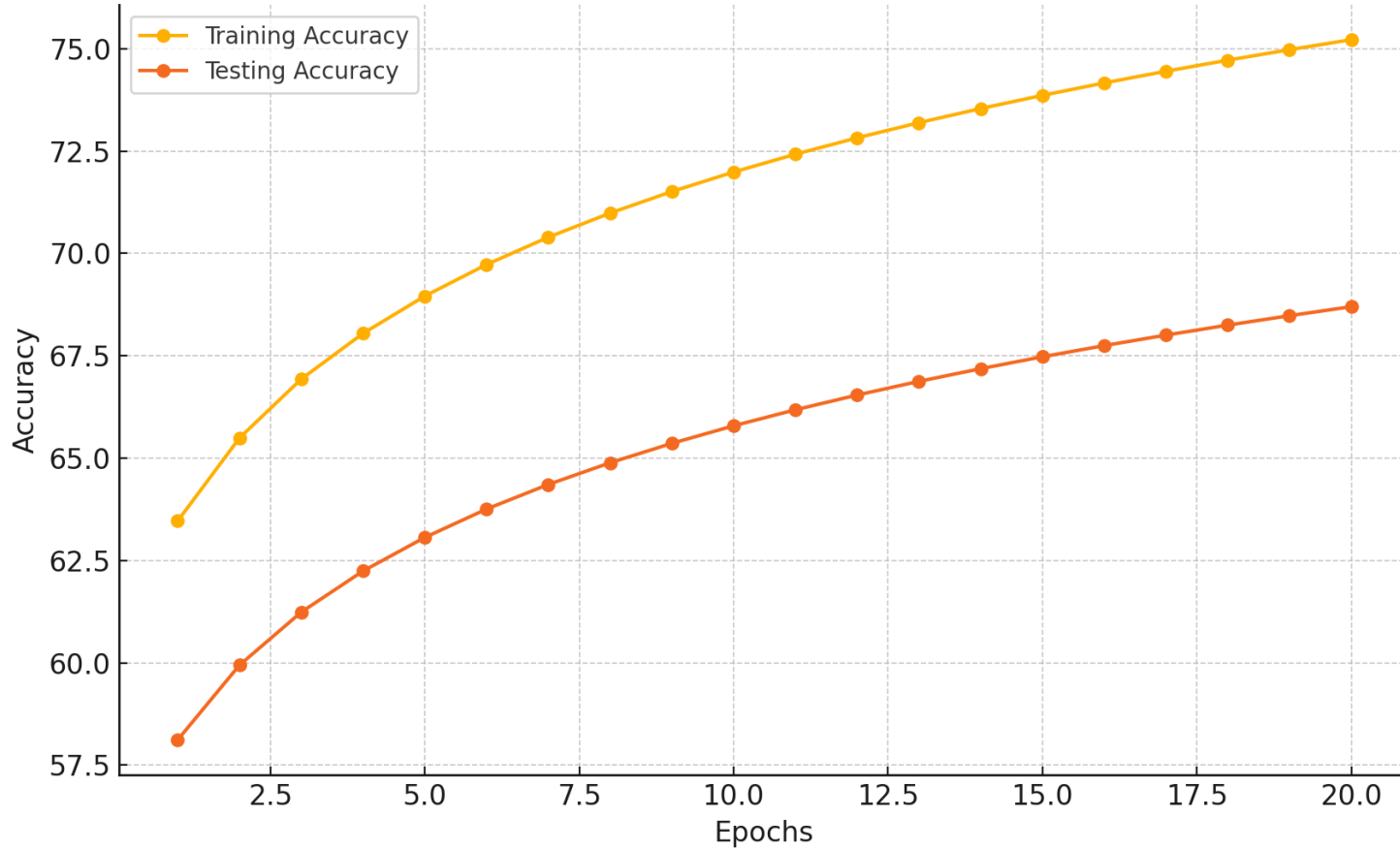
# Testing Data

- **Key Points:**
- Testing uses data the model has not seen during training.
- It helps assess the model's ability to perform on new data.
- A good model should have similar performance on both training and test data.

- Training Data:
- Larger portion of the dataset is typically allocated for training (e.g., 70-80%).
- Testing Data:
- Smaller portion of the dataset is typically reserved for testing (e.g., 20-30%).



# Training vs Testing Performance



# Overfitting

## **Definition:**

Overfitting occurs when a model learns the training data too well, including noise and specific patterns, resulting in poor generalization to new data.

# Overfitting

- **Symptoms: High** accuracy on **training data** but **poor** performance on **testing data**.
- **Causes:**
  - - Too complex model.
  - - Insufficient training data.
- Example: A decision tree with too many branches fitting every single data point.

- *Question:* You train a model that achieves 99% accuracy on training data but only 70% on test data. What is likely happening here?

- *Answer:* The model is overfitting to the training data. It's learned specific details that don't generalize to new data, leading to poor performance on the test set.

# Underfitting

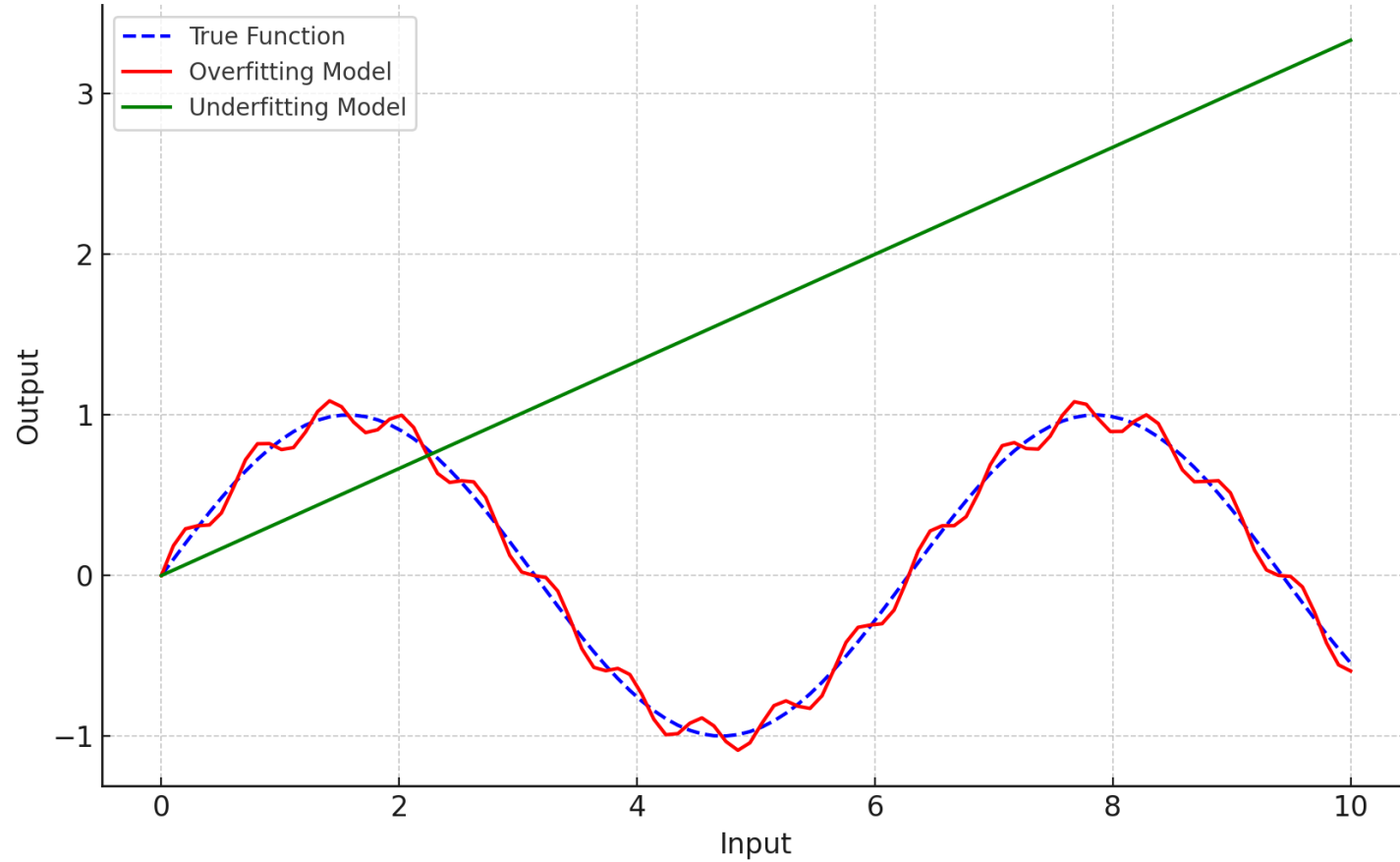
- **Definition:** Underfitting occurs when a model is **too simple** to capture the underlying patterns in **the data**.
- Symptoms: **Poor** performance on both **training** and **testing data**.
- Causes:
  - - Model is too simple.
  - - Insufficient training time.
- Example: A linear regression model applied to non-linear data.

- *Question:* You have a linear model trained on complex data, and it achieves only 50% accuracy on both training and test sets. What could be the issue?

- *Answer:* The model is underfitting; it's too simple to capture the complexity of the data. A more complex model might improve performance.



## Overfitting vs Underfitting



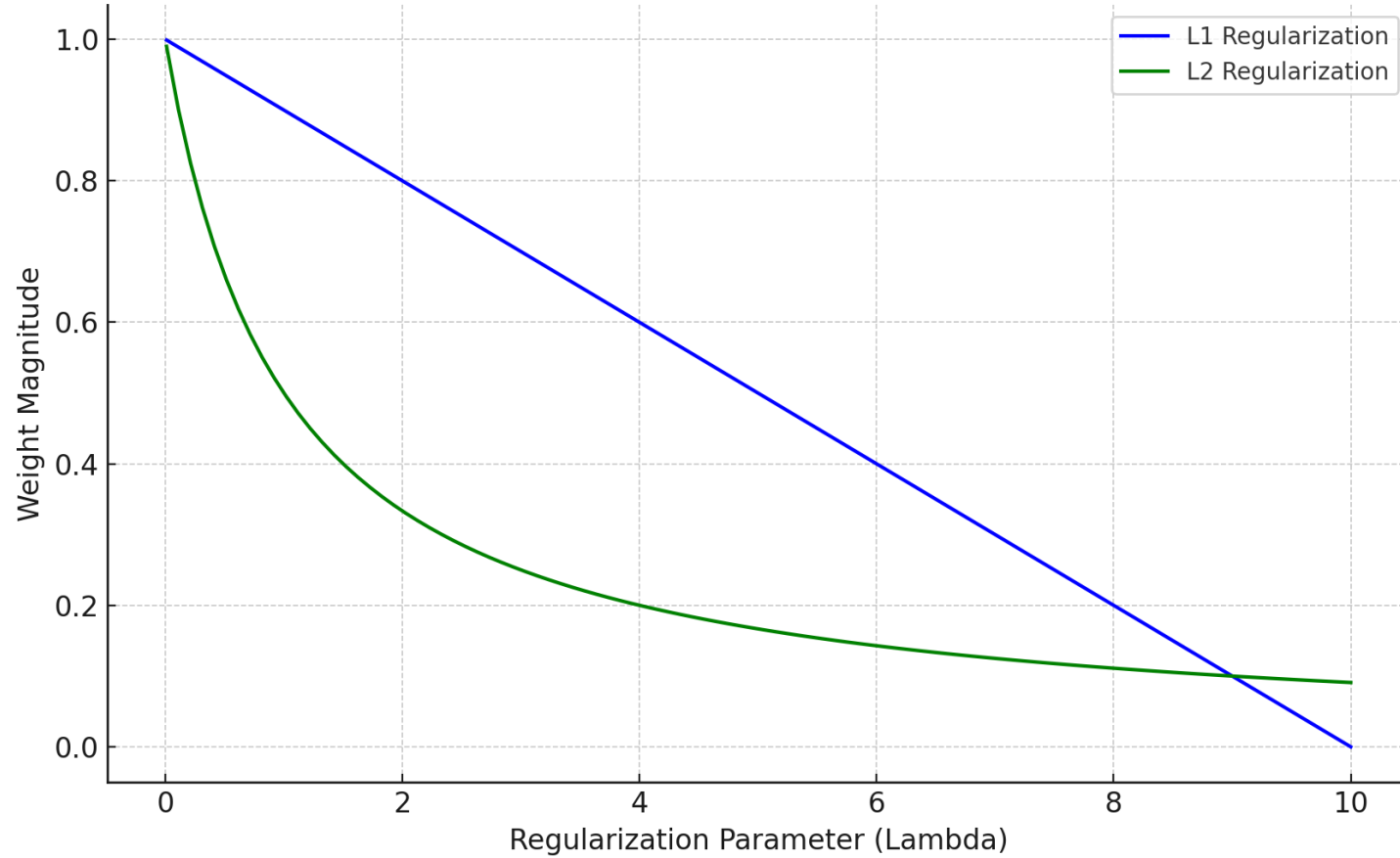
# Regularization

- Definition: Regularization techniques are used to prevent overfitting by adding a penalty to the loss function.
- Types:
  - - L1 Regularization (Lasso): Adds absolute value of coefficients as penalty.
  - - L2 Regularization (Ridge): Adds squared value of coefficients as penalty.
- Process:
  - 1. Apply regularization during model training.
  - 2. Adjust regularization parameter to balance bias and variance.
- Example: Logistic regression with L2 regularization to prevent overfitting.

# Comparing Overfitting and Underfitting

- **Overfitting:**
  - - Complex model.
  - - High variance.
  - - Low bias.
  - - Poor generalization.
- **Underfitting:**
  - - Simple model.
  - - Low variance.
  - - High bias.
  - - Poor generalization.
- Visualization: Graphs showing overfitting and underfitting.

Effect of Regularization on Weights



# Regularization Techniques

- L1 Regularization: Encourages sparsity (many zero coefficients).
- L2 Regularization: Distributes error across all terms.
- Elastic Net: Combines L1 and L2 regularization.
- Dropout (in neural networks): Randomly drops neurons during training to prevent co-adaptation.
- Example: Regularization applied to neural network training.

# Conclusion

- Summary of key points:
  - - Importance of training and testing data.
  - - Avoiding overfitting and underfitting.
  - - Utilizing regularization to improve model performance.
- Encouragement to explore further.
- Q&A session.